4 MINOR EDITS, 301-399

CARDS 300, PLOT TIME SPAN

This card is optional for NEW and RESTART problems, are required for PC Window Plotting Case. If not entered, the initial time in CARD 2XX is set as a minimum value and final time is set as a maximum value.

W1(R) minimum X-value for X-Y Plot.

W2(R) Maximum X-Value for X-Y Plot.

CARDS 301 THROUGH 399, MINOR EDIT DATA REQUESTS

These cards are optional for NEW and RESTART problems, are required for a REEDIT problem, and are not allowed for PLOT and STRIP problems. If these cards are not present, no minor edits are printed. If these cards are present, minor edits are generated, and the order of the printed quantities is given by the card number of the request card. One request is entered per card, and the card numbers need not be consecutive. For RESTART problems, if these cards are entered, all the cards from the previous problem are deleted.

W1(A) Variable code (alphanumeric).

W2(I) Parameter (numeric).

W3(R) Minimum Y Value for X-Y Plot (Optional Select for Window Plotting).

W4(R) Maximum Y Value for X-Y Plot (Optional Select for Window Plotting).

W5(I) Window Identification for multiple X-Y Plot.

W6(I) Color Pallet Identification for multiple X-Y Plot.

Words 1 and 2 form the variable request code pair. The quantities that can be edited and the input required are listed below. For convenience, quantities that can be used in plotting requests, in trip specifications, as search variables in tables, and as operands in control statements are listed. Units for the quantities are also given. Interactive input variables described in Section 6 can be used in batch or interactive jobs in the same manner as the variables listed below. The parameter for interactive input variables is 1000000000. Quantities compared in variable trips must have the same units, and input to tables specified by variable request codes must have the specified units. The quantities are listed in alphabetical order within each section.

The underlined quantities without an asterisk in Section 4.1 through Section 4.8 are always written to the restart-plot file (RSTPLT). Underlined quantities followed by an asterisk have only some of the quantities written to the restart-plot file and the text will indicate which quantities are written.

4.1 General Quantities

Variable	Description
COUNT	Current attempted advancement count number. The parameter is 0.
CPUTIME	Current CPU time for this problem (s). The parameter is 0.
DT	Current time step (s). The parameter is 0.
DTCRNT	Current Courant time step (s). The parameter is 0.
EMASS	Estimate of mass error in all the systems (kg, lb). The parameter is 0.
ERRMAX	Current estimate of the truncation mass error fraction. The parameter is 0. This is the maximum of the two types of computed mass error (ε_m or ε_{rms}).
EXTSNN	Extra system variables where NN goes from 01 to 20. These variables are output to the restart-plot file when RELAP5 is built using the "extra" flag on the dinstls command line. These variables are useful for debugging the code and for making temporary changes to the code. There is a similar set of variables for volumes called EXTVNN and for junctions called EXTJNN. Parameter is system number N.
NULL	Specifies null field. Allowed only on trip cards. The parameter is 0.
RKTPOW3D	Total reactor power for the MARS/MASTER kinetics coupled code (W). It is the sum of the fission and decay powers (W) for all the core heat structures. This variable is written to the restart-plot file only when card 1 option 88 is used.
STDTRN	Steady-state/transient flag. The parameter is 0. For steady-state, the value is 0.0. For transient, the value is 1.0.
SYSERG	Total energy in system N (kJ, Btu). Parameter is system number N.
SYSMER	Estimate of mass error in system n (kg, lb). Parameter is system number N.
SYSTMS	Total mass of steam, water, and noncondensable in system N (kg, lb). Parameter is system number N.

TESTDA An array testda, of twenty quantities, [REAL TESTDA(20)] has been defined for the convenience of program developers. This entry with a parameter ranging from 1 through 20 selects TESTDA(parameter). The testda array is initially set to 0.0, and programming must be inserted to set TESTDA values. The usual purpose of this capability is to allow a simple method for debug information to be printed in minor edits or to be plotted.

TIME Time (s). The parameter is 0. This request cannot be used for minor edit requests.

TIMEOF Time of trip occurring (s). The parameter is the trip number. This request is allowed only on trip cards.

<u>TMASS</u> Total mass of water, steam, and noncondensables in all the systems (kg, lb). The parameter is 0.

4.2 Component Quantities

The quantities listed below are unique to certain components; for example, a pump velocity can only be requested for a pump component. The parameter is the component number, i.e., the three-digit number CCC used in the input cards.

Variable	Description
ACPGTG	Accumulator vapor specific heat, C_p, at vapor temperature (J/kg•K, Btu/lb•^{o}F).
ACPNIT	Accumulator noncondensable specific heat, C_p, at vapor temperature (J/kg•K, Btu/lb•^{o}F).
<u>ACQTANK</u>	Total energy transport to the gas by heat and mass transfer in the accumulator (W, Btu/s).
<u>ACRHON</u>	Accumulator noncondensable density (kg/m^3, lb/ft^3).
<u>ACTTANK</u>	Mean accumulator tank wall metal temperature (K, ^{o}F).
<u>ACVDM</u>	Gas volume in the accumulator tank, standpipe, and surge line (m^3, ft^3).
ACVGTG	Accumulator vapor specific heat, C_v, at vapor temperature (J/kg•K, Btu/lb•^{o}F).
<u>ACVLIQ</u>	Liquid volume in the accumulator tank, standipipe, and surge line (m^3, ft^3).
AHFGTF	Accumulator heat of vaporization at liquid temperature (J/kg, Btu/lb).
AHFGTG	Accumulator heat of vaporization at vapor temperature (J/kg, Btu/lb).

AHFTG	Accumulator liquid enthalpy at vapor temperature (J/kg, Btu/lb).
AHGTF	Accumulator vapor enthalpy at liquid temperature (J/kg, Btu/lb).
AVGTG	Accumulator specific volume at vapor temperature (m^3/kg, ft^3/lb).
AVISCN	Accumulator noncondensable viscosity (kg/m•s, lb/ft•s).
BETAV	Accumulator steam saturation coefficient of expansion (K^{-1}, $^oF^{-1}$).
CDIM	GE mechanistic dryer critical inlet moisture quality.
DIM	GE mechanistic dryer inlet moisture quality.
DMGDT	Accumulator/time rate of change in dome vapor mass (kg/s, lb/s).
GDRY	GE mechanistic separator capacity factor.
OMEGA	Inertial valve disk angular velocity (rad/s, rev/min).
<u>PMPHEAD</u>	Pump head in the pump component (Pa, lb_f/ in^2).
PMPMT	Pump motor torque (N•m, lb_f•ft).
PMPNRT	Calculated pump inertia (kg•m ^{2}, lb•ft^2).
<u>PMPTRQ</u>	Pump torque in the pump component (N•m, lb_f•ft).
<u>PMPVEL</u>	Pump rotational velocity in the pump component (rad/s, rev/min).
PRZLVL	Pressurizer level in the PRIZER component (m, ft).
THETA	Inertial valve disk angular position (deg).
<u>TUREFF</u>	Efficiency of the turbine component.
<u>TURPOW</u>	Power developed in the turbine component (W, Btu/s).
<u>TURTRQ</u>	Torque developed in the turbine component (N•m, lb_f•ft).
<u>TURVEL</u>	Rotational velocity of the turbine component (rad/s, rev/min).

VLVAREA Ratio of the current valve physical area to the junction area. The junction area is the fully open valve physical area for the smooth area option and the minimum of the two connecting volumes for the abrupt area change.

VLVSTEM Ratio of the current valve stem position to the fully open valve stem position for the motor and servo valves when the normalized stem position option is used. For the motor and servo valves when the normalized area option is used and for all the other valves, this is the ratio of the current valve physical area to the fully open valve physical area.

XCO GE mechanistic separator liquid carryover quality.

XCU GE mechanistic separator vapor carryunder quality.

XI GE mechanistic separator inlet quality.

4.3 Volume Quantities

For most of the following variable codes, the parameter is the volume number, i.e., the nine-digit number CCCNN0000 printed in the major edit. The parameter is CCC010000 for a single-volume; CCC010000 for a time-dependent volume; CCCNN0000 for a volume in a pipe component ($01 \leq NN \leq 99$); CCC010000 for the volume in a branch, separator, jetmixer, turbine, or ECC mixer component; CCC010000 for the volume in a pump component; and CCC010000 for the volume in an accumulator component. Some of the quantities are associated with the coordinate directions in the volume, and these quantities are computed for each coordinate direction in use. The parameter for the coordinate direction-related quantities is the volume number plus F, where F is described below. The quantities requiring the volume number plus F are so identified.

Every volume has at least one coordinate direction, and some volumes may have up to three orthogonal coordinate directions. Each coordinate has an inlet face and an outlet face. Faces are numbered 1 through 6, where faces 1 and 2 are the inlet and outlet faces associated with coordinate 1 (or x), respectively, faces 3 and 4 are inlet and outlet faces associated with coordinate 2 (or y), and faces 5 and 6 are inlet and outlet faces associated with coordinate 3 (or z). All volumes use coordinate 1. The quantity F to be added to the volume number to form the parameter used with coordinate direction related quantities is 0 or the face number. When F is 0 (i.e., just the volume number) or 1 or 2, the volume velocity is for coordinate 1. When F is 3 or 4, the volume velocity is for coordinate 2, and when F is 5 or 6, the volume velocity is for coordinate 3. For the underlined quantities followed by an asterisk in the list below, the coordinate-dependent quantities for coordinate 1 are automatically written to the restart-plot records using the parameter with F equal to 0. Input checks are made to ensure the parameter specifies a volume coordinate direction that is in use.

Variable	Description
AVOL	Area of the volume (m^2, ft^2); the parameter is the volume number plus F.

BETAFF Liquid isobaric coefficient of the thermal expansion, β_f, bulk conditions (K^{-1}, $^oF^{-1}$).

BETAGG Vapor isobaric coefficient of the thermal expansion, β_g, bulk conditions (K^{-1}, $^oF^{-1}$).

BORON Spatial boron density, ρ_b (kg/m^3, lb/ft^3). This is the volume liquid fraction (α_f) times the liquid density (ρ_f) times the boron concentration (C_b). Boron concentration is used for hydrodynamic input, and boron density is used for minor edits and plots.

BRNPPM Boron concentration in ppm unit.

CSUBPF Liquid specific heat, C_{pf}, bulk conditions (J/kg•K, Btu/lb•^{o}F).

CSUBPG Vapor specific heat, C_{pg}, bulk conditions (J/kg•K, Btu/lb•^{o}F).

DRFDP Partial derivative of ρ_f with respect to P (s^2/m^2, s^2/ft^2).

DRFDUF Partial derivative of ρ_f with respect to U_f ($kg{\bullet}s^2/m^5$, $lb{\bullet}s^2/ft^5$).

DRGDP Partial derivative of ρ_g with respect to P (s^2/m^2, s^2/ft^2).

DRGDUG Partial derivative of ρ_g with respect to U_g ($kg{\bullet}s^2/m^5$, $lb{\bullet}s^2/ft^5$).

DRGDXA Partial derivative of ρ_g with respect to X_n (kg/m^3, lb/ft^3).

DTDP Partial derivative of T^s with respect to P (K/Pa, $in^2{\bullet}^oF/lb_f$).

DTDUG Partial derivative of T^s with respect to U_g ($s^2{\bullet}K/m^2$, $s^2{\bullet}^oF/ft^2$).

DTDXA Partial derivative of T^s with respect to X_n (K, ^{o}F).

DTFDP Partial derivative of T_f with respect to pressure (K/Pa, $in^2{\bullet}^oF/lb_f$).

DTFDUF Partial derivative of T_f with respect to U_f ($s^2{\bullet}K/m^2$, $s^2{\bullet}^oF/ft^2$).

DTGDP Partial derivative of T_g with respect to P (K/Pa, $in^2{\bullet}^oF/lb_f$).

DTGDUG Partial derivative of T_g with respect to U_g ($s^2{\bullet}K/m^2$, $s^2{\bullet}^oF/ft^2$).

DTGDXA Partial derivative of T_g with respect to X_n (K, ^oF).

EXTVNN Extra volume variables where NN goes from 01 to 20. These variables are output to the restart-plot file when RELAP5 is built using the "extra" flag on the dinstls command line. These variables are useful for debugging the code and for making temporary changes to the code. There is a similar set of variables for junctions called EXTJNN and for systems called EXTSNN.

FLOREG Flow regime number; the parameter is the volume number. A chart showing the meaning of each number is shown in Volume I, Section 3.1.3.

FWALF Liquid wall frictional drag coefficient (kg/m^3•s, lb/ft^3•s); the parameter is the volume number plus F.

FWALG Vapor wall frictional drag coefficient (kg/m^3•s, lb/ft^3•s); the parameter is the volume number plus F.

GAMMAC For explicit coupling of heat conduction/transfer and hydrodynamics, this is 0. For implicit coupling of heat conduction/transfer and hydrodynamics, this is the mass transfer rate per unit volume at the vapor/liquid interface in the boundary layer near the wall for condensation (kg/m^3•s, lb/ft^3•s).

GAMMAI Mass transfer rate per unit volume at the vapor/liquid interface in the bulk fluid for vapor generation/condensation (kg/m^3•s, lb/ft^3•s).

GAMMAW For explicit coupling of heat conduction/transfer and hydrodynamics, this is the mass transfer rate per unit volume at the vapor/liquid interface in the boundary layer near the wall for vapor generation/condensation (kg/m^3•s, lb/ft^3•s). For implicit coupling of heat conduction/transfer and hydrodynamics, this is the mass transfer rate per unit volume at the vapor/liquid interface in the boundary layer near the wall for vapor generation (kg/m^3•s, lb/ft^3•s).

HGF Direct heating heat transfer coefficient per unit volume (W/m^3•K, Btu/s•ft^3•^oF).

HIF Liquid side interfacial heat transfer coefficient per unit volume (W/m^3•K, Btu/s•ft^3•^oF).

HIG Vapor side interfacial heat transfer coefficient per unit volume (W/m^3•K, Btu/s•ft^3•^oF).

HSTEAM Steam specific enthalpy at bulk conditions using partial pressure of steam (J/kg, Btu/lb).

HVMIX Enthalpy of the liquid and vapor (J/kg, Btu/lb).

P Volume pressure (Pa, lb_f/in^2).

PECLTV Peclet number.

PPS Steam partial pressure (Pa, lb_f/in^2).

Q Total volume heat source from the wall and direct moderator heating to liquid and vapor (W, Btu/s). This variable request is the same as "Q.wall.tot." in the major edits.

QUALA Volume noncondensable mass fraction. The ratio of the mass of the noncondensable gas to the total mass of the vapor phase.

QUALE Volume equilibrium quality. This quality uses phasic enthalpies and mixture quality, with the mixture enthalpy calculated using the flow quality.

QUALS Volume static quality.

QWG Volume heat source from the wall and direct moderator heating to vapor (W, Btu/s). This variable request is the same as "Qwg.wall.gas." in the major edits.

RHO Total density (kg/m^3, lb/ft^3).

RHOF Liquid density ρ_f (kg/m^3, lb/ft^3).

RHOG Vapor density ρ_g (kg/m^3, lb/ft^3).

RHOM Total density for the mass error check (kg/m^3, lb/ft^3).

SATHF Liquid specific enthalpy at saturation conditions using partial pressure of steam (J/kg, Btu/lb).

SATHG Steam specific enthalpy at saturation conditions using partial pressure of steam (J/kg, Btu/lb).

SATTEMP Volume saturation temperature based on the partial pressure of steam (K, ^oF).

SIGMA Surface tension (N/m, lb_f/ft).

SOUNDE Volume sonic velocity (m/s, ft/s).

TEMPF Volume liquid temperature T_f (K, ^oF).

TEMPG Volume vapor temperature T_g (K, ^oF).

TFSUB Degree of subcooling for volume liquid (K, ^oF), <0.0.

TGSUP Degree of superheating for volume vapor (K. ^oF), >0.0.

THCONF Liquid thermal conductivity (W/m•K, Btu/s•ft•^oF).

THCONG Vapor thermal conductivity (W/m•K, Btu/s•ft •^oF).

TIENGV Total internal energy (of both phases and noncondensables) in volume (J, Btu).

TMASSV Total mass (includes both phases and noncondensables) in volume (kg, lb).

TSATT Saturation temperature corresponding to total pressure (K, ^oF).

UF Liquid specific internal energy (J/kg, Btu/lb).

UG Vapor specific internal energy (J/kg, Btu/lb).

VAPGEN Total mass transfer rate per unit volume at the vapor/liquid interface in the bulk fluid for vapor generation/condensation and in the boundary layer near the wall for vapor generation/condensation (kg/m^3•s, lb/ft^3•s).

VELF* Volume oriented liquid velocity (m/s, ft/s); the parameter is the volume number plus F.

VELG* Volume oriented vapor velocity (m/s, ft/s); the parameter is the volume number plus F.

VISCF Liquid viscosity (kg/m•s, lb/ft•s).

VISCG Vapor viscosity (kg/m•s, lb/ft•s).

VOIDF Volume liquid fraction.

VOIDG Volume vapor fraction (void fraction).

VOIDLA Void above the level.

VOIDLB Void below the level.

VOLLEV Location of the level inside the volume (m, ft).

VVOL Volume of the volume (m^3, ft^3).

4.4 Junction Quantities

For the following variable request codes, the parameter is the junction number, i.e., the nine-digit number CCCNN0000 printed in the major edit. The parameter is CCC000000 for a single-junction; CCC000000 for a time-dependent junction; CCCMM0000 for a junction in a pipe component (01 ≤ MM ≤ 99); CCCMM0000 for a junction in a branch, separator, jetmixer, turbine, or ECC mixer component (01 < MM ≤ 9); CCC000000 for a valve junction; CCC010000 for the inlet junction in a pump component; CCC020000 for the outlet junction in a pump component; CCCIINN00 for a junction in the multiple-junction component (01 ≤ II ≤ 99, 01 ≤ NN ≤ 99); and CCC010000 for the junction in an accumulator component.

Variable	Description
C0J	Junction distribution coefficient. The 0 in C0J is the number 0 and not the upper case letter O.
CHOKEF	Junction choking flag. The value is 0 if the flow is not choked, and is 1 if the flow is choked.
EXTJNN	Extra junction variables where NN goes from 01 to 20. These variables are output to the restart-plot file when RELAP5 is built using the “extra” flag on the dinstls command line. These variables are useful for debugging the code and for making temporary changes to the code. There is a similar set of variables for volumes called EXTVNN and for systems called EXTSNN.
FIJ	Interphase friction coefficient ($N \bullet s^2/m^5$, $lb_f \bullet s^2/ft^5$).
FJUNFT	Total forward user input form loss coefficient for irreversible losses, including Re dependence (dimensionless).
FJUNRT	Total reverse user input form loss coefficient for irreversible losses, including Re dependence (dimensionless).
FLENTH	Total enthalpy flow in junction (includes both phases and noncondensables) (J/s, Btu/s).
FLORGJ	Junction flow regime number. A chart showing the meaning of each number is shown in Volume I, Section 3.
FORMFJ	Liquid abrupt area change model form loss factor (dimensionless).
FORMGJ	Vapor abrupt area change model form loss factor (dimensionless).
FWALFJ	Non-dimensional liquid wall friction coefficient (dimensionless).

FWALGJ Non-dimensional vapor wall friction coefficient (dimensionless).

IREGJ Vertical bubbly/slug flow junction flow regime number. A chart showing the meaning of each number is shown in Volume I, Section 3.

MFLOWJ Combined liquid and vapor flow rate (kg/s, lb/s).

MFLOWFJ Liquid flow rate (kg/s, lb/s).

MFLOWGJ Vapor flow rate (kg/s, lb/s).

MFLOWNJ Noncondensible gas flow rate (kg/s, lb/s).

QUALAJ Junction noncondensable mass fraction.

RHOFJ Junction liquid density (kg/m^3, lb/ft^3).

RHOGJ Junction vapor density (kg/m^3, lb/ft^3).

SONICJ Junction sound speed (m/s, ft/s). This speed is based on the physical junction area. It does not include the effects of the throat ratio and the discharge coefficients.

UFJ Junction liquid specific internal energy (J/kg, Btu/lb).

UGJ Junction vapor specific internal energy (J/kg, Btu/lb).

VELFJ Junction liquid velocity (m/s, ft/s). This velocity is based on the junction area A_j.

VELGJ Junction vapor velocity (m/s, ft/s). This velocity is based on the junction area A_j.

VGJJ Vapor drift velocity (m/s, ft/s).

VOIDFJ Junction liquid fraction.

VOIDGJ Junction vapor fraction (void fraction).

VOIDJ Junction vapor fraction (void fraction) used in the interphase friction.

XEJ Junction equilibrium quality.

4.5 Heat Structure Quantities

For the request code, HTVAT, the parameter is the seven-digit heat structure number CCCG0NN. For the remaining codes, the parameter is the seven-digit heat structure number CCCG0NN with a two-digit number appended. For codes other than HTTEMP and HTVAT, the appended number is 00 for the left boundary and 01 for the right boundary. For HTTEMP, the appended number is the mesh point number. For HTVAT, omit the two appended digits and use only the seven digit number. Only the left and right surface temperatures are written by default in plot records on the RSTPLT file, and thus, plot requests in plot-type problems and strip requests are limited to those temperatures

Variable	Description
GAPPRES	Gap pressure (Pa).
GAPCOND	Gap effective conductivity (W/m-K, Btu/s•ft-R)
HTCHF	Critical (maximum) heat flux (W/m^2, Btu/s•ft^2).
HTCHFR	Critical hat flux ratio (ratio of HTCHF to HTRNR). This variable can only be plotted and is set equal to the constant value of 12.34567 whenever HTCHF or HTRNR is equal to 0.0.
HTGAMW	Wall vapor generation rate per unit volume (kg/m^3•s, lb/ft^3•s). The parameter is the heat structure geometry number CCCG0NN with a two-digit number appended (00 for the left boundary, and 01 for the right boundary).
HTHTC	Heat transfer coefficient (W/m^2•K, Btu/s-ft^2•^{o}F).
HTMODE	Boundary heat transfer mode number (unitless). The mode number indicates which heat transfer regime is currently in effect. The parameter is the seven-digit heat structure geometry number, CCCG0NN, with a two-digit number appended. The two-digit appended number, 00, specifies the left boundary, and 01 specifies the right boundary. This same quantity is valid for the reflood heat structures.
HTRG	Heat flux to vapor phase (W/m^2, Btu/s•ft^2). The parameter is the heat structure geometry number, CCCG0NN, with a two-digit number appended (00 for the left boundary, and 01 for the right boundary).
HTRNR	Heat flux (W/m^2, Btu/s•ft^2).
HTTEMP	Mesh point temperature (K, ^{o}F). The parameter is the heat structure geometry number CCCG0NN with a two-digit number appended (mesh point number). The surface

temperatures are written to the plot record but interior mesh point temperatures must be requested through the 2080XXXX cards.

HTVAT	Volume averaged temperature in the heat structure (K, ^oF).
PECL	Liquid Peclet number for the heat structures. The parameter is the heat structure geometry number CCCG0NN with a two-digit number appended (00 for the left boundary, and 01 for the right boundary).
QRAD	Heat flux (W/m^2, Btu/s.ft^2) due to radiation and contact heat transfer. The parameter is the heat structure geometry number, CCCG0NN, with a two-digit number appended (00 for the left boundary, and 01 for the right boundary).
STANT	Stanton number. The parameter is the heat structure geometry number CCCG0NN with a two-digit number appended (00 for the left boundary, and 01 for the right boundary).

4.6 Reflood-Related Quantities

For the following variable codes, the parameter is the heat structure geometry number, i.e., the four-digit number CCCG printed in the major edit.

Variable	Description
FINES	Current number of axial nodes on a reflood structure.
TCHFQF	Temperature at the critical (maximum) heat flux (K, ^oF).
TREWET	Rewet, quench, Leidenfrost, or minimum film boiling temperature (K, ^oF).
ZQBOT	Elevation of bottom quench front (m, ft).
ZQTOP	Elevation of top quench front (m, ft).

4.7 Reactor Kinetic Quantities

The parameter is 0 for the following reactor kinetic quantities.

Variable	Description
REAC	Reactivity feedback total (dollars). This is the sum of REACM, REACRB, REACS, and REACTF.
RKBORN	Reactivity feedback from boron density changes (dollars).

RKDOPP Reactivity feedback from fuel temperature changes (dollars).

RKMOD Reactivity feedback total from moderator (dollars). This is the sum of RKMODD and RKMODT.

RKMODD Reactivity feedback from moderator density changes (dollars).

RKMODT Reactivity feedback from moderator temperature (spectral) changes (dollars).

RKSCRAM Reactivity feedback from scram curve (dollars).

RKFIPOW Reactor power from fission (W).

RKGAPOW Reactor power from decay of fission products and actinides (W).

RKPOWA Reactor power from decay of actinides (W).

RKPOWK Reactor power from decay of fission products (W).

RKREAC Reactivity (dollars).

RKRECPER Reciprocal period (s^{-1}).

RKTPOW Total reactor power, i.e., sum of fission and decay powers (W).

4.8 Control System Quantities

The parameter is the control component number, i.e., the three-digit number, CCC, or the four-digit number, CCCC, used in the input cards.

CNTRLVAR Control component number. These quantities are assumed dimensionless except for a SHAFT component.